KARMANYA RAINA

Mumbai | 8355864166 | rainakarmanya7@gmail.com

EDUCATION

College of Engineering Pune (COEP)

B.Tech Electrical Engineering (2024–2028) | CGPA: 7.00

Atomic Energy Central School–4, Mumbai — CBSE Class X: 87.6%

PACE Jr. College, Nerul — Maharashtra Board, Class XII: 74.17%

EXPERIENCE

NPCIL Summer Intern | June–July 2026

- Built a signal processing pipeline in MATLAB to analyze and condition RTD sensor data, improving reliability of temperature measurement readouts.
- Modeled and tuned digital filter parameters (order, cutoff, sampling rate) through iterative simulation to optimize noise rejection for real-world sensor data.
- Validated system performance using zero-phase processing techniques and quantitative metrics (RMSE, tolerance-based checks), ensuring measurement accuracy against reference data.
- Explored digital logic design fundamentals on FPGA, including counters and shift registers, as an introduction to hardware-level system design.

PROJECTS

PID Control System Simulations | MATLAB, Simulink (https://github.com/karmanya0712/PID-Control-System-Simulations.git)

- Modeled a DC motor using its transfer function in MATLAB/Simulink and tuned a PID controller to study transient response characteristics such as overshoot, settling time, and steady-state error.

Digital Lock using Logic Gates

- Designed and implemented a combinational digital security circuit from basic logic gates, translating a functional access-control requirement into gate-level logic.

TECHNICAL SKILLS

Engineering Tools: MATLAB, Simulink

Core Competencies: System Modelling & Simulation, Signal Processing, Sensor Data Analysis, Performance Validation, Algorithm Development

Technical Domains: Embedded Systems, Digital Electronics, Control Systems, FPGA Fundamentals

CERTIFICATIONS

- MathWorks — MATLAB Onramp

ACHIEVEMENTS

Represented India at Junior World Cup Bridge (France, 2016)